

Bridging Actionability Gap in Online How-To Instruction-Following Through Human-Chatbot Interaction

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ABSTRACT

People frequently come across unfamiliar tasks to complete in daily and professional life. A common strategy to accomplish such tasks is to refer to online How-To tutorials (e.g., wikiHow pages) and follow the instructions step-by-step. However, such archived tutorials are not always actionable, since the sequence of actions is pre-defined and could result in a discrepancy between what was shared and what people need in operational situations. Differing from regular conversational instruction-giving, where people are able to communicate instructions adaptively based on the ongoing working status, archived procedures lack such flexibility. As such, people may face actionability issues in which they may have difficulties navigating the instruction, making decisions on what to do, and solving problems. My dissertation explores system designs using chatbots to accompany novice users to improve the situational actionability and social acceptability of online tutorials for non-expert users. I present ActionaBot, an experimental platform for exploring human-chatbot collaborative instruction giving to augment online tutorials. By participating in the Doctoral Consortium, I am eager to receive feedback from experienced CSCW researchers to inform my research direction. In the long run, I aim to contribute a better understanding of how people and the chatbot may collaborate to enable actionable knowledge sharing and transfer at a large scale.

CCS CONCEPTS

• **Human-centered computing** → *Collaborative and social computing systems and tools*; **HCI design and evaluation methods**; *Empirical studies in HCI*.

KEYWORDS

How-To tutorials, chatbot, instruction following, situated action, reflection-in-action

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1 INTRODUCTION AND BACKGROUND

People are in need to accomplish familiar and unfamiliar tasks in everyday life and work. When handling tasks one's expertise and experience can't match, online resources of archived tutorials and instructions become potentially relevant and useful. Meanwhile, people have needs to keep documentation to record and explain the procedures of how to achieve the desired goal through a series of operational steps ([14, 15]) to other people. For instance, how to fix a disposal, how to bake a Chiffon cake, etc. Such procedural instructions that lead people to transform the status of work from one state to another to achieve the personally desirable outcome are commonly referred as "How-Tos" or "Know-Hows". Similar to how Wikipedia was constructed and grown as storage of encyclopedic knowledge, people also started to construct, share and deposit How-To instructions through various platforms, such as wikiHow¹, Instructables² and eHow³.

While online tutorials increase the availability and accessibility of How-Tos knowledge covering various topics, simply depositing tutorials online doesn't necessarily address the actionability aspect of needs. People may still not be able to follow the sequence of actions explicated in the instruction. The limitation in actionability issue may origin from two aspects. First, there could be a communication gap between those who initially shared the how-to and those who want to reenact it. Authors who wrote instructions could have the skills to accomplish the corresponding task, but they may lack the ability to communicate the steps clearly in their expositions. For instance, they might overlook recording a step unknown to novices but very commonly known by experts; they may use jargon without sufficient explanation; they may not consider enough of novices' perspectives and don't decompose a step into feasible granularity for reenactment ([7, 8, 12, 16]). Second, novices following a pre-archived tutorial may face a new situation different from the old situation associated with the original tutorial. For instance, depending on their ongoing work states, they may not be able to make inferences or take actions that are not considered in the tutorial; they may have trouble solving problems happening in the execution and could not move on to the next step; or they have difficulties to freely navigate the tutorial and could not go back on track after going back to review the previous step([3, 4]). Therefore, we are curious to investigate if novices could be benefited by receiving guidance in an in-situation and on-demand way when following the instruction to finish a task, so that they could dynamically adjust the plan to take action at each step based on the

¹<https://www.wikihow.com/Main-Page>, Access Date: 03.01.2022

²<https://www.instructables.com/about/>, Access Date: 03.01.2022

³<https://www.ehow.com/about>, Access Date: 03.01.2022

current situation of execution, including the prerequisite to take action, ability to take action and the action outcome.

Situational planning for problem-solving is powerful and used in face-to-face instruction giving and following through conversation. People take turns to share and acquire work- and collaboration-related information, such as the states of the joint work (e.g., “which step are we in?”) and co-workers (e.g., “have you finished that step?”) and the work plan (e.g., “shall we move to the next step?”). Through conversations, team members are able to explicitly coordinate their actions and achieve synergies ([6]). However, the situational plan mechanism is not available in the archived online instruction, where the sequence of actions is pre-planned without considering the individual novice’s actual circumstances during the instruction-following.

To introduce situational guidance into How-Tos at scale, I propose to incorporate a chatbot system, namely ActionaBot, into online How-To instruction giving/following to provide human-chatbot conversational support for tutorial navigation and in-situ planning, and help novice workers complete the tasks. Beyond designing the content and structure of chatbot interactions, I also explore how to make the chat more sociable and acceptable for novices. Following previous work in CSCW indicates that giving instruction with higher emotion would increase social acceptability in human-human interaction [1, 11] and we know less about it in human-chatbot interaction, it would be interesting to change emotion cues in ActionaBot’s feedback and observe novices’ acceptability.

I ultimately aim to improve the ‘actionability’ of online tutorials for public. I want to answer the following research questions:

R1: Can ActionaBot mediate novices to follow online archived How-Tos by providing situational feedback on scale?

R2: How to leverage ActionaBot to augment novices’ ability to follow unfamiliar operational instructions? when and how to prompt users to adjust the action plan towards actionability and acquisition of know-hows?

R3: Is ActionaBot’s situational feedback more acceptable when conveying higher emotional expressiveness?

To illustrate my research, I introduce my ongoing project called ActionaBot in section 2.1 and a finished project about incorporating emotional expressiveness in chatbot’s response in section 2.2. Meanwhile, I introduce the plan for upcoming studies and expected contributions.

2 WORK IN PROGRESS AND EXPECTED CONTRIBUTION

2.1 ActionaBot: human-chatbot conversation is feasible and helpful for novices while following online How-To instruction

I started by understanding novices’ behaviors and needs during instruction-following and whether we could leverage chatbot to augment the conversational interaction between the user and archived instruction tutorial on scale. To this end, I designed and implemented a chatbot system, ActionaBot, that is embedded in the same page as step-by-step instruction. A unified interaction space was provided for users to read instructions and interact with ActionaBot.

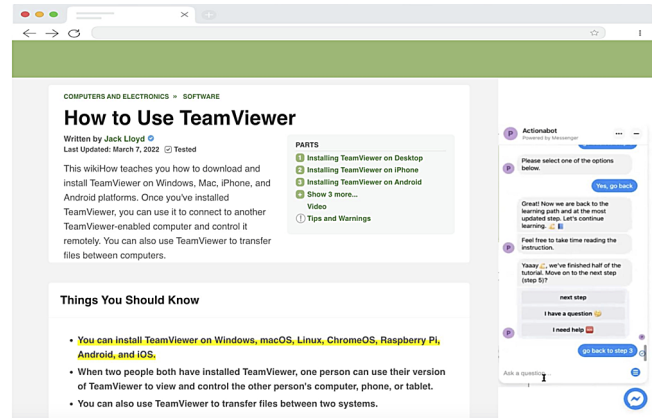


Figure 1: ActionaBot in a How-To tutorial

The chatbot design follows the principle that ActionaBot needs to reach mutual awareness on working states and mutual affirmation on the next step to do with users by eliciting user’s understanding of working states, disclosing chatbot’s understanding of user’s working states, reaching agreement on the current working state, and confirm the next step. The chatbot is also able to recognize users’ intents and responses properly, especially when the user is not able to move on, the bot identifies the inactionability step and collaborates with users to resolve the next step through conversation. To have the same context with users, the chatbot keeps the parsed tutorial as a knowledge base and maintains a view of the user trace indicating when the user is on which step.

Then, I conducted an initial online study in 2021 to check ActionaBot’s effects on supporting novices to navigate the tutorial and novices’ reaction to ActionaBot. I invited people with limited knowledge in programming and software development to work with ActionaBot on an advanced unfamiliar software installation task (Install Python 3 on a remote computer in real-time). Participants were asked to share their screens and I used fly-on-the-wall ([2]) to take observation notes, with minimum intervention. After finishing the task, a 30-minute semi-structured interview was conducted to better understand participants’ experiences and thought processes during the study. I found: (1) novices although might perceive instructions to be inactionable due to prior experience and how instructions were authored, they were able to follow conversational guidance and willing to adapt to the chatbot through turn-taking to calibrate working states back-and-forth. (2) users’ awareness of the work status increased with the conversational prompts from the chatbot. However, how different novices converse with the chatbot to follow instructions from independent follow the instruction is still unclear.

Currently, I’m conducting a mix-methods study to compare two instructions following conditions – with and without ActionaBot (Fig.1). Following instruction without ActionaBot could be modeled as following pre-planned action sequences, where the type and order of actions are fixed before novices do the task. On the other hand, following instructions with ActionaBot could be treated as a situational-plan, where the plan for each step is collaboratively discussed by the user and ActionaBot and adjusted according to the

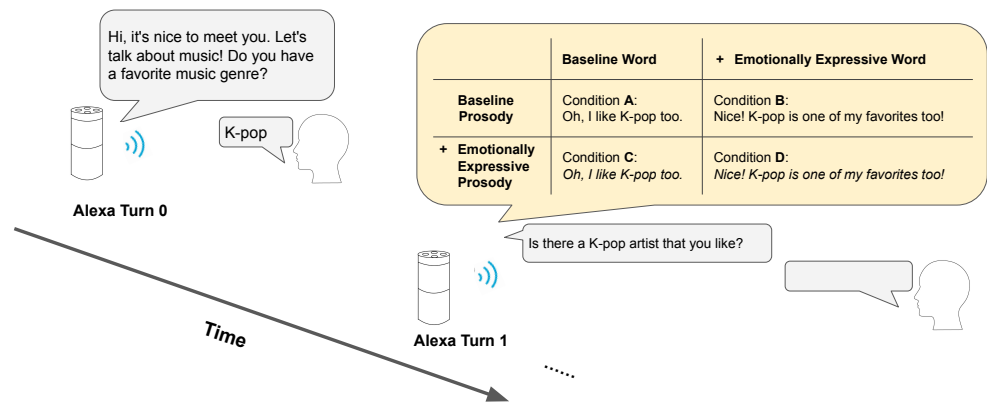


Figure 2: Schematic for the experimental conditions in Chatbot Emotional Acknowledgement Study. In responding to a user’s response, the chatbot produces lexical and prosodic emotional expressiveness across four conditions in the acknowledgment.

current situation. I hypothesize that novices could potentially think and generate ideas about what they are doing while the execution (so-called reflection-in-action [13]). Therefore novices reach higher self-awareness and feel the instruction is more actionable. I’m collecting both qualitative data (from observation of users’ behavior and retrospective interviews) and quantitative (from user trace data, such as user’s intent transition timeline, the number and length of utterances, and time spent on each step to test the hypothesis) to better understand the role of ActionaBot. I want to contribute empirical understandings of the actionability gap for novices in How-To instruction following and derive design implications on “when and how to give what guidance” by ActionaBot.

2.2 Social Acceptability of Situational feedback with Emotion Cues

Apart from incorporating situational feedback mechanism into ActionaBot, another underexplored aspect is how to convey guidance so that novices are more willing to accept the instruction. My thesis is going to explore how emotional expressiveness conveyed in the ActionaBot’s feedback would affect novices’ perception and the reaction of How-To instruction actionability.

Previous work [5, 9, 10] reflects divergent effects of emotion cues on instruction following and learning. Lin et al. and Park and Lim indicated that learning outcomes increased when using emotional visual illustration in the tutoring system, while Foster et al. pointed out that using a virtual robot to give emotional feedback makes users feel increased difficulty to answer questions in instruction-following task. In my finished project on understanding how users perceive more emotionally expressive acknowledgment (Fig. 2) from the voice assistant chatbot in the music-related social chat, I conducted a two-pronged approach, including a user interaction study (Experiment 1 – between-subjects design) and an independent perception study (Experiment 2 – within-subjects design). I found: (1) Both studies provide converging evidence that increasing emotional prosody yields more positive perceptions

of chatbot interactions, in increasing perception of emotional expressiveness (Experiments 1 and 2) as well as overall engagement, human-likeness, and likability of the bot (Experiment 2); (2) The proximity between the user and chatbot matters. Direct users are significantly influenced by both prosody and word expressiveness, while by-stand users are only significantly influenced by prosody emotional expressiveness.

My thesis is in a stage where I extend the emotional expressiveness dimension from prosodic-level and word-level to multimedia-level, where I will include GIFs to convey positive emotions as an emotional communication strategy to cheer up, encourage and praise novices for various events happening during the instruction following. The plan is to further focus on understanding how and when to incorporate emotional expressive feedback into ActionaBot and exploring whether emotional factor help or bother novices’ acceptance of instructions and working state communication with ActionaBot. According to the take-away from the emotional acknowledgment project, I will focus on the direct interaction between users and ActionaBot with two emotional expressiveness factors: multimedia GIF and prosody. A well-controlled 2*2 lab study will be conducted with a data collection of user behavior trace and retrospective interview. I would like to contribute more insights on the social acceptability of situational instruction given by chatbot counterpart.

In summary, with above mentioned two main studies, I want to push the frontier of human-chatbot interaction and explore the possibility of human-chatbot collaboration in the online procedural knowledge-sharing process. The ultimate goal would be enabling novice-chatbot collaboration on following procedural instructions and overcoming inactionability issues. I would like to derive design implications through an experimental study on how to give scalable and actionable guidance by leveraging the chatbot and how to design such chatbot to enhance novice workers’ abilities.

3 MOTIVATION TO ATTEND CSCW DOCTORAL CONSORTIUM

Possessing a background from both NLU and HCI, I'm interested in interdisciplinary and want to explore how to conduct human-centric research with benefits from advanced technology. In my dissertation, I'm taking a novel approach to informing the chatbot system design by borrowing insights from human-human collaboration. From the Doctoral Consortium, I hope to conduct discussions with peers and senior CSCW researchers on the approach used in my dissertation and gain valuable feedback on the research scope, research questions and methodologies presented. I also want to calibrate my research with the expectations of the community. Due to the COVID-19, a new research schema is introduced and new challenges are introduced to establish connections with other researchers and study participants, Doctoral Consortium provides me a precious opportunity to participate as a part of CSCW, get connect with other researchers, and learn actionable research experience from others experienced researchers.

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